

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (CL) Examination

April 2013

CL 406 Design of Structures - II

Date: 25.04.2013, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. IS 456:2000, IS 800:2007, IS 875:1987, SP -16, Steel Table are allowed.

SECTION - I

Q - 1 (a) Design a continuous reinforced concrete beam of rectangular section to support dead load of 8 kN/m and live load of 10 kN/m over 3 spans of 5 m each. The ends are simply supported. Take M 20 grade concrete and Fe 415 steel. Sketch the details of reinforcements for the beam. [07]

Q - 2 (a) Design the longitudinal steel for a braced column 400×300 mm bent in single curvature with the following data: [07]

Effective length about both axes = 6.0 m, unsupported length = 6.5 m, factored load = 2000 kN, factored moments on major axis are 230 kNm at the top and 170 kNm at bottom. The factored moments about minor axis are 100 kNm at top and 80 kNm at the bottom. Take M 35 grade concrete and Fe 415 steel.

(b) Design a rectangular combined footing connecting two columns A and B, 3.5 m centre to centre, carrying an ultimate axial load of 800 kN and 1200 kN respectively. The boundary line of the property is 380 mm from the outer face of the column A. Column A is $350 \text{ mm} \times 350 \text{ mm}$ and column B is $400 \text{ mm} \times 400 \text{ mm}$. The bearing capacity of the soil obtained from plate load test is 100 kN/m^2 . Use M 20 grade concrete and Fe 415 steel. Assume effective cover equal to 60 mm. [07]

OR

Q - 2 (a) Determine the reinforcement required for a braced column against side sway with the following data: [07]

Size of the column = $300 \times 400 \text{ mm}$, effective lengths l_{ex} and l_{ey} = 6.0 and 5.0 m, respectively; unsupported length $l = 7 \text{ m}$, factored load $P_u = 1400 \text{ kN}$, factored moments in the direction of larger dimension = 60 kNm at top and 25 kNm at bottom, factored moments in the direction of shorter dimension = 50 kNm at top and 25 kNm

at bottom. The column is bent in double curvature. Reinforcement will be distributed equally on four sides. Take M 30 grade concrete and Fe 415 steel.

- (b) Design a trapezoidal footing for the two columns A and B transmitting service loads of 700 kN and 1400 kN respectively. The column A is 350 mm square and column B is 550 mm square in size and they are spaced at 4.5 m centers. The property line is 280 mm beyond the face of column A. The safe bearing capacity of soil at site is 140 kN/m². Use M 20 grade concrete and Fe 415 steel. [07]

Q - 3 (a) A counterfort retaining is required to suit the following data: [07]

Height of wall above ground level = 5 m

SBC of soil = 150 kN/m²

Angle of internal friction $\phi = 30^\circ$

Density of soil = 15 kN/m³

Spacing of counterforts = 2.8 m c/c

Coefficient of friction between soil and concrete = 0.42

Adopt M 20 grade concrete and Fe 415 HYSD bars.

Design the vertical stem and do all necessary checks.

- (b) A staircase slab spanning longitudinally is set into pockets of two supporting beams, one of either end, in successive landings between the floors. If the effective span of the slab is 3 m horizontally and the rise of the stair is 1.5 m with 250 mm treads and 150 mm risers, design and detail the staircase slab. Assume a live load of 3 kN/m², finishes 0.5 kN/m², M 20 grade concrete and Fe 415 HYSD bars. [07]

OR

- Q - 3 (a) A cantilever retaining wall is used to retain earth 4.0 m above ground level. The density of soil is 15 kN/m³ and its angle of repose is 30°. The embankment is horizontal at top. The safe bearing capacity of soil is 180 kN/m² and the coefficient of friction between soil and concrete is 0.4. Surcharge pressure is 16 kN/m². Adopt M 20 grade concrete and Fe 415 HYSD bars. Design the heel slab and do all necessary checks. [07]

- (b) A longitudinal type of staircase spans a distance of 3.75 m centre-to-centre of beams. The rise $R = 175$ mm, going $G = 250$ mm and tread $T = 270$ mm. The treads have 15 mm granolithic finish and consist of 15 steps. Assuming grade of concrete as M 25 and grade of steel as Fe 415, design the staircase for a live load of 5 kN/m². Assume breadth of staircase as 1.5m. [07]

SECTION – II

Q - 4 (a) A column ISHB 350 @ 661.2 N/m carries an axial compressive factored load of 1700 kN. Design a suitable bolted gusset base. The base rests on M 15 grade concrete pedestal. Use 24 mm diameter bolts of grade 4.6 for making the connections. [07]

Q - 5 (a) Calculate dead load, live load and wind load per panel point for the roof truss shown in Figure 1 situated in the vicinity of Pune. [05]

Spacing of truss = 4 m, Span of truss = 9 m, Slope of truss = 30° ,

Height of eaves = 6 m, Roofing material = G.I. Sheets, $k_1 = 1$, $k_2 = 0.981$, $k_3 = 1$

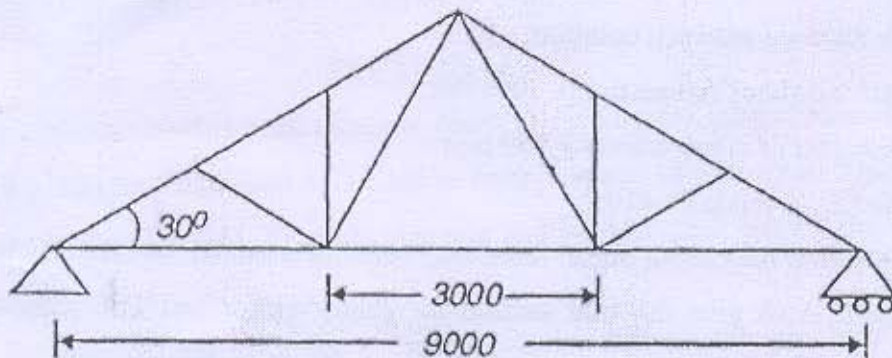


Figure: 1

(b) A welded plate girder 24 m in span and laterally restrained throughout. It has to support a uniform load of 100 kN/m throughout the span exclusive of self-weight. Design the girder without intermediate transverse stiffeners. The steel for the flanges and web plates is of grade Fe 410. Design the cross section and do the checks for bending strength and shear capacity of web. [09]

OR

(b) Design the welded plate girder for an effective span of 30 m and carrying a uniformly distributed load of 30 kN/m and two concentrated loads of 150 kN each acting at 10 m from both ends. The girder is simply supported at ends. It is fully restrained at both ends against lateral buckling throughout the span. The steel for the flanges and web plates is of grade Fe 410. Design the cross section and do the checks for bending strength and shear capacity of web. [09]

Q - 6 (a) Design a purlin on a sloping roof truss for the following data: [05]

Spacing of truss = 4 m, Spacing of purlins = 1.73 m, Angle of roof truss = 30°

Dead Load = 0.375 kN/m

Live Load = 0.6 kN/m

Wind Load (Suction) = 1.3 kN/m

Use Channel section as a purlin. Do the check for moment capacity.

- (b) A gantry girder is to be used in an industrial building carrying a manually operated [09]
overhead travelling crane, for the following data:

Crane capacity – 200 kN

Self-weight of the crane girder excluding trolley – 200 kN

Self-weight of the trolley, electric motor, hook etc. – 40 kN

Approximate minimum approach of the crane hook to the gantry girder – 1.20 m

Wheel base – 3.5 m

c/c distance between gantry rails – 16 m

c/c distance between columns – 8 m

Self-weight of rail section – 300 N/m

Diameter of crane wheels – 150 mm

Steel is of grade Fe 410.

Calculate maximum shear force and bending moment for vertical loads and lateral loads. Also give the trial section for gantry girder and find moment of inertia of gantry girder.

OR

- (b) A gantry girder section is shown in **Figure 2**. The girder is subjected to following [09]
moments:

Moment about major axis, $M_z = 650$ kNm

Moment about minor axis, $M_y = 20$ kNm

Check the adequacy of the girder for combined local capacity.

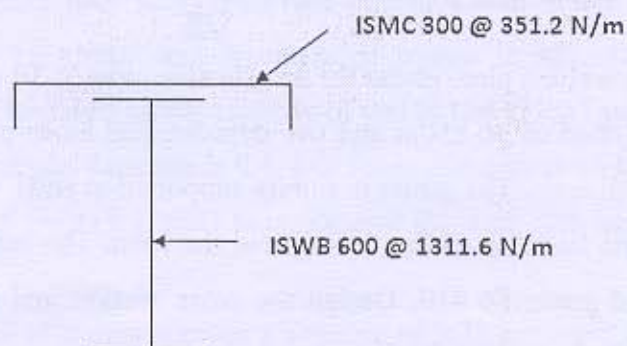


Figure: 2
